

Transport Poverty Index

A replicable multi-dimensional and multi-modal measure of transport poverty

Alice Anonymous^{a,*}, Bob Security^a, Cat Memes^a, Filipe Moura

^a*CERIS, Instituto Superior Técnico - Universidade de Lisboa, Av Rovisco Pais 1, 1049-001
Lisbon, Portugal*

Abstract

This is the abstract. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Vestibulum augue turpis, dictum non malesuada a, volutpat eget velit. Nam placerat turpis purus, eu tristique ex tincidunt et. Mauris sed augue eget turpis ultrices tincidunt. Sed et mi in leo porta egestas. Aliquam non laoreet velit. Nunc quis ex vitae eros aliquet auctor nec ac libero. Duis laoreet sapien eu mi luctus, in bibendum leo molestie. Sed hendrerit diam diam, ac dapibus nisl volutpat vitae. Aliquam bibendum varius libero, eu efficitur justo rutrum at. Sed at tempus elit.

Keywords: keyword1, keyword2

Please make sure that your manuscript follows the guidelines in the Guide for Authors of the relevant journal. It is not necessary to typeset your manuscript in exactly the same way as an article, unless you are submitting to a camera-ready copy (CRC) journal.

For detailed instructions regarding the elsevier article class, see <https://www.elsevier.com/authors/policies-and-guidelines/latex-instructions>

*Corresponding author

Email addresses: `alice@example.com` (Alice Anonymous), `bob@example.com` (Bob Security), `cat@example.com` (Cat Memes), `fmoura@tecnico.pt` (Filipe Moura)

1. Heading 1

Here are two sample references: [Pereira et al. \(2021\)](#) [INE \(2018\)](#). If citing the other way ([Lovelace et al., 2022](#)).

1.1. Heading 2

If `cite-method` is set to `citeproc` in `elsevier_article()`, then `pandoc` is used for citations instead of `natbib`. In this case, the `cs1` option is used to format the references. By default, this template will provide an appropriate style, but alternative `cs1` files are available from <https://www.zotero.org/styles?q=elsevier>. These can be downloaded and stored locally, or the url can be used as in the example header.

2. Methods

2.1. Normalisation

All indicators are normalised to the range $[1, 100]$ using min-max scaling:

Benefit indicators (higher raw value $\rightarrow$ better):

$$X_{norm} = 1 + \frac{X - X_{min}}{X_{max} - X_{min}} \times 99$$

Cost indicators (lower raw value $\rightarrow$ better):

$$X_{norm} = 1 + \frac{X_{max} - X}{X_{max} - X_{min}} \times 99$$

- **Accessibility** metrics (`access_*`) are *benefit* indicators.
- **Mobility** cost metrics (`mobility_cost_*`), PT waiting times, all **affordability** indicators, and all **safety** indicators are *cost* indicators.

Missing values in accessibility indicators (arising from infeasible PT routes) are imputed with the mean of the respective variable prior to PCA.

2.2. PCA-Based Dimension Scores

For each dimension (Accessibility, Mobility, Safety) and at the global (all-mode) and per-mode levels, a PCA is run using `FactoMineR::PCA()` at parish level.

The **first principal component (PC1)** score is used as the dimension index, after rescaling to $[1, 100]$. PC1 captures the dominant axis of variation among the normalised indicators.

For **Affordability**, PCA is not applied (too few indicators); instead, the normalised composite `h_transp_inc_comp` (housing + transport cost as share of income, modal-share weighted) is used directly as the Affordability Index.

For **Safety per mode**, an equal-weight average of the severity index (per mode) and total vehicles involved (per mode) is used instead of PCA.

For **Mobility - Car** PCA is not applied (too few indicators); instead, we assumed an equal-weight average of the normalized composite `mobility_commuting_avg_tt_car` and `road_length`.

2.3. Composite IMPT Scores

Three aggregation methods are computed for the global IMPT and each per-mode IMPT:

1. **Geometric mean** (equal weights, Figure 1a):

$$IMPT_{geom} = \left(\prod_d (S_d + 1) \right)^{1/n} - 1$$

Partially compensatory: penalizes dimensions with very low values non-linearly. Highlights the most critical situations, where at least one dimension is severely deficient.

2. **Arithmetic mean** (equal weights, Figure 1b):

$$IMPT_{avg} = \frac{1}{n} \sum_d S_d$$

Compensatory: high vulnerability in one dimension can be offset by good performance in others. Suitable for setting aggregate policy targets and communicating to non-specialist audiences.

3. Entropy-weighted geometric mean (Figure 1c):

$$IMPT_{entropy} = \prod_d (S_d + 1)^{w_d} - 1$$

where weights w_d are derived from the **entropy weighting method**:

Weights are determined by the empirical dispersion of the dimensions across parishes: the dimension that varies most among territories receives greater weight. Maximises territorial contrast and is useful for competitive prioritisation of interventions.

$$w_d = \frac{d_d}{\sum_j d_j}, \quad d_d = 1 - e_d, \quad e_d = -k \sum_i p_{id} \ln(p_{id})$$

with $k = 1/\ln(n)$ and p_{id} being the proportion of the dimension score for area i .

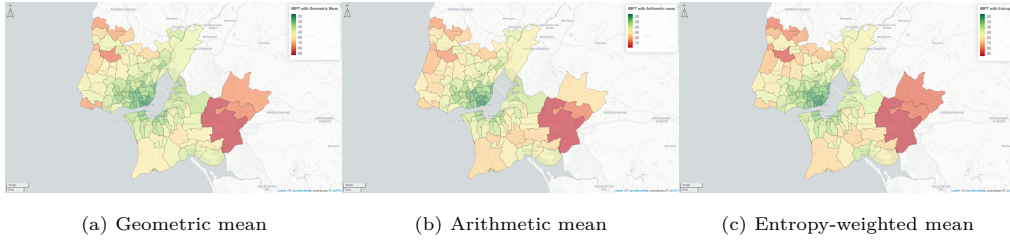


Figure 1: IMPT results for different aggregation methods for all modes (with Navegante)

3. Figures and tables

Figure 2 is generated using an R chunk.

4. Tables coming from R

Tables can also be generated using R chunks, as shown in Table 1 example.

```
knitr::kable(head(mtcars)[,1:4])
```

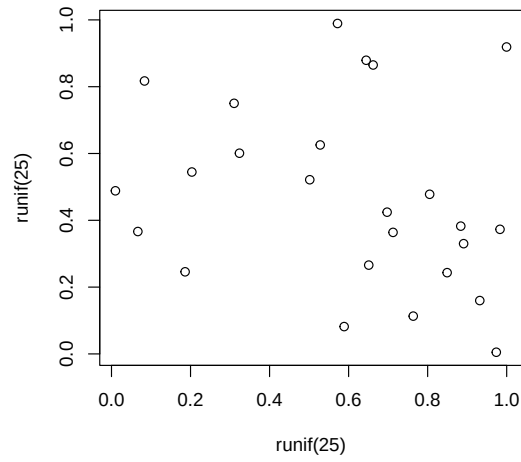


Figure 2: A meaningless scatterplot

Table 1: Caption centered above table

	mpg	cyl	disp	hp
Mazda RX4	21.0	6	160	110
Mazda RX4 Wag	21.0	6	160	110
Datsun 710	22.8	4	108	93
Hornet 4 Drive	21.4	6	258	110
Hornet Sportabout	18.7	8	360	175
Valiant	18.1	6	225	105

References

- INE, I.N.d.E., 2018. Inquérito à mobilidade nas Áreas metropolitanas do porto e de lisboa — imob 2017. URL: <https://www.ine.pt/xurl/pub/349495406>.
- Lovelace, R., Félix, R., Carlino, D., 2022. Jittering: A computationally efficient method for generating realistic route networks from origin-destination data. Findings doi:[10.32866/001c.33873](https://doi.org/10.32866/001c.33873).
- Pereira, R.H.M., Saraiva, M., Herszenhut, D., Braga, C.K.V., Conway, M.W., 2021. r5r:

Rapid realistic routing on multimodal transport networks with R5 in R. Findings
doi:[10.32866/001c.21262](https://doi.org/10.32866/001c.21262).